

Labwork 3.2 - Pattern Drawing Machine

General Information and Submission Rules:

- In this labwork, you need to design a machine that prints different patterns based on the user choice.
- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork3.c**. In addition to this, you need to create a **command line history** file with this command:
history > history_fullname_labwork3.txt
and submit this also.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (15p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to the Pattern Drawing Machine! By using this machine you can do:
1. Draw Rectangle
2. Draw Square
3. Draw Triangle
4. Diamond Pattern
5. Exit
Make your choice:
```

Question 2 - Draw Rectangle (15p) and Square (15p)

- Create new functions (**draw_rectangle** and **draw_square**) that displays rectangle and square shapes by using asterisks (*).
- **draw_rectangle** function will take two parameters from the user (height and width).
- Since it is not a square, check that the height and width are not equal to each other then draw the pattern to the user.
- On the other hand, the **draw_square** function will take one parameter (edge_length) since each edge is equal with each other.

```
Make your choice: 1
Enter height and width: 4 10
*****
*****
*****
*****
Make your choice: 2
Enter the edge length: 2
**
**
Make your choice:
```

Question 3 - Draw Triangle (25p)

- Create a new function (**draw_triangle**) that displays triangle shape by using asterisks (*).
- It will take the (base) as a parameter from the user and then draw the pattern to the user.
- If the shape cannot be drawn, print an error to the user and then ask the base again.

```
Make your choice: 3
Enter the base: 6
The base must be an odd number!
Enter the base: 5
*
***
*****
Enter the choice:
```

Question 4 - Diamond Pattern (25p) and Exit (5)

- Create a new function (**diamond**) that displays diamond shape by using asterisks (*).
- It will take the (base) as a parameter from the user and then draw the pattern to the user.
- If the shape cannot be drawn, print an error to the user and then ask the base again.

```
Enter the choice: 4
Enter the base: 6
The base must be an odd number!
Enter the base: 9
  *
 ***
*****
*****
*****
*****
  ***
   *

Enter the choice: 5
Exiting the program. Good bye!
```